

Q1 (12 points)

Using careful measurements, you have determined subsequent ionization energies (in kJ/mol) for element X given below:

688.3(I_{E1})

1457.3(I_{E2})

2928.9(I_{E3})

4244.5(I_{E4})

17213.3(I_{E5})

20109.2 (I_{E6})

- Which main group (in the periodic table) would you place element X in? Explain your reasoning.
- Using the ionization energy data for element X given above, determine the longest wavelength (in nm) of a photon required to remove an electron from the outer "ns" orbital of the appropriate X cation.

Q2 (12 points)

a. Two Chem110 students have decided to make a sensor using metal A, metal B, and metal C. They want to construct a sensor using a light source such that only two of the three metal surfaces eject electrons. The work function for metal A is 4.96 eV, for metal B is 5.97 eV, and for metal C is 3.82 eV. $1\text{eV} = 1.602 \times 10^{-19}\text{ J}$

Student 1 decides to use 275 nm wavelength light source, and Student 2 decides to use 225 nm wavelength light source. Which of the two students (1 or 2) is more likely to construct a successful sensor? Explain your answer and show calculations.

- What will be the velocity (m/s; three significant figures) of an ejected electron of metal C (work function = 3.82 eV), if a light source of 400 kJ/mol is used?

Q3 (10 points)

a.

Name the orbital, for Ca atom (Assume ground state electronic configuration)

(I) Orbital containing the electron that requires the maximum energy to be removed.

Name of the orbital:

(II) Orbital containing the electron that experiences the minimum effective nuclear charge.

Name of the orbital:

For the ground state Ca atom, draw the radial probability distribution for the above orbitals on one plot

b.

In a single atom:

(i) How many electrons can have the following two quantum numbers? $n = 5; m_l = -2$

(ii) How many electrons can have the following two quantum numbers? $n = 9; l = 2$

Q4:

For each molecule below, draw the most stable Lewis structure showing appropriate bond angles/shape and dashed/wedged bond where appropriate. Show lone pairs.

 H_5IO_6 No O-O bonds

 CH_3NH_2

b. For the *balanced* reaction: $\text{C}_3\text{H}_4(g) + 4\text{O}_2(g) \longrightarrow 3\text{CO}_2(g) + 2\text{H}_2\text{O}(g)$

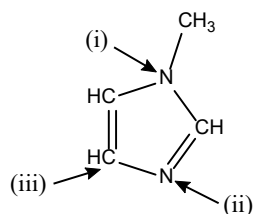
Calculate (in kJ) the standard enthalpy change (ΔH) for the combustion of 20.0 grams of propyne (C_3H_4). Give your answer to 3 significant figures.

Bond:	O=O	$\text{C}\equiv\text{C}$	C-C	C-H	O-H	C=O
Bond energy (kJ/mol):	347	839	346	413	467	799

c. Which of the following has a higher lattice energy? Explain your answer

KCl or CsI

5.



For the molecule and indicated atoms (denoted with an arrow, (i) to (iii)) answer the following questions. Consider the structure as given (no need to consider resonance structures). All atoms have formal charge of 0 (lone pairs are not shown). (Use valence bond theory and hybridization to describe the bonding.)

a) How many **total** sp^2 -s sigma bonds are in the given molecule

b) 1. Describe the orbitals involved in all bonds formed by N (i). Designate sigma/pi bonds.

2. What is the electron **and** molecular geometry at N (i)?

3. Circle the one value that best describes the C-N-C bond angle at N(i).

>90° to <109.5° 109.5° >109.5° to <120° 120° >120° to <180

c) 1. Describe the orbitals involved in all bonds formed by N (ii). Designate sigma/pi bonds. **(2 points)**

2. What is the molecular geometry at N (ii). **(1 point)**

3. Circle the one value that best describes the C-N-C bond angle at N (ii). **(1 point)**

>90° to <109.5° 109.5° >109.5° to <120° 120°

d) What is the electron **and** molecular geometry at (iii) **(2 points)**

Q6:

Arrange the following. Briefly explain your answer

1) In increasing order (smallest to largest) of bond angles

1. H_2S
2. XeF_2
3. $[SbF_6]^-$

2) In increasing order (smallest to largest) of size

1. Br^-
2. PO^{2-}
3. K^+
4. Sr^{2+}

3) In increasing order (smallest to largest) of bond length of CN bond in:

1. CH_3NH_2
2. HCN
3. CH_3CHNH

4) In increasing order (smallest to largest) of number of unpaired electrons:

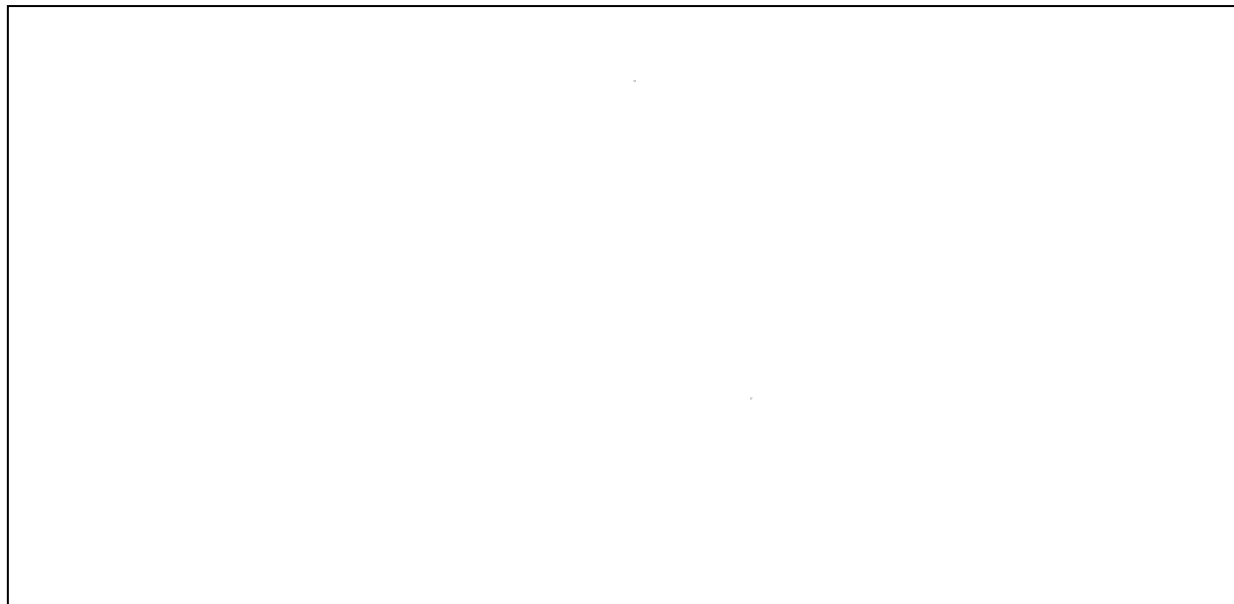
1. Sc^+
2. Cr^+
3. As
4. Au^{2+}

Q7 (14 points)

- a. In atomic spectra of H, a visible emission line is observed at 434 nm. Determine the level (n_{initial}) from which the transition occurred
- b. Frodo is walking fast towards Mount Doom. What is the minimum uncertainty in his position, given that his velocity is $1000 \pm 2.0 \text{ mm} \cdot \text{s}^{-1}$? Frodo has a mass of 50 kg.

Question 8

- a. Draw the complete MO diagram (valence orbitals only) for the molecule CN. Make sure to label atomic and molecular orbitals. Label the energy axis. What is the bond order?



b.

Which of the following will require a higher energy to ionize (assume removal of easiest to remove electron)? Explain your answer

C, N, CN